Granular Instrumental Variables

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The Law of Large Numbers fails in modern economies

In a standard economy, micro-shocks cancel out at a rate of $1/\sqrt{N}$ Note.

In a **granular** economy, where the size distribution of firms follows a power law, a few firms are so large that their idiosyncratic luck drives aggregate volatility.

Endogenous aggregate shocks confound the effect of large firms

The Goal: Estimate the causal elasticity of wages to labor demand (ψ).

The Identification Challenge: **Simultaneity**

- **Supply Equation:** Wages respond to labor demand and macro supply shocks.
- **Demand Equation:** Labor demand responds to wages and macro demand shocks.

The Simple Model

Difference-in-averages as an instrument

A simple model of aggregate and idiosyncratic shocks

Consider an economy with N firms.

The system is given by:

$$w_t = \psi L_{St} + \varepsilon_t \tag{Supply}$$

$$L_{it} = \phi^d w_t + \eta_t + u_{it} \tag{Demand}$$

Where:

- w_t is the wage
- $L_{St} = \sum_i S_i L_{it}$ is the size-weighted aggregate outcome (number of workers)
- η_t and ε_t are aggregate shocks
- ψ is the labor supply elasticity, ϕ^d is the labor demand elasticity
- u_{it} is the idiosyncratic shock to firm i

OLS fails because aggregate wages and common shocks are correlated

We cannot estimate ψ and ϕ^d by OLS.

Why?

- ε_t and η_t are typically correlated.
- This implies that L_{St} is correlated with ε_t in the supply equation.
- And w_t is correlated with η_t in the demand equation.

GIV estimates elasticities via Two-Stage Least Squares

First Stage: Isolate exogenous variation in labor demand

$$L_{St} = \beta z_t + \text{Controls} + \nu_t$$

We regress aggregate labor demand (L_{St}) on the Granular Instrument (z_t). This extracts $\widehat{L}_{St}$, the variation driven *only* by the idiosyncratic shocks of superstar firms.

Second Stage: Estimate the causal elasticity

$$w_t = \psi \widehat{L}_{St} + \text{Controls} + \varepsilon_t$$

We regress aggregate wages (w_t) on the predicted labor demand from the First Stage ($\widehat{L}_{St}$) to obtain an unbiased estimate of ψ.

Validating the Instrument: Relevance and Exclusion

For z_t to be a valid instrument, two fundamental assumptions must hold in a granular economy.

- **Relevance (Cov$(L_{St}, z_t) \neq 0$):**

  The granular shocks must actually move the aggregate market. This requires a *concentrated market* (e.g., high Herfindahl index) where the idiosyncratic shocks of the largest firms are volatile enough to shift total labor demand.

- **Exclusion Restriction (Cov$(z_t, \varepsilon_t) = 0$):**

  The idiosyncratic shocks of large firms (z_t) must not directly affect aggregate wages (w_t) through any channel other than aggregate labor demand (L_{St}). They must be truly orthogonal to macro aggregate supply shocks (ε_t).

Constructing the Granular Instrumental Variable (z_t)

The GIV exploits the difference between size-weighted and equal-weighted averages.

Define the equal-weighted average outcome:

$$L_{Et} = \frac{1}{N} \sum_{i=1}^{N} L_{it} \qquad \text{(Equal-Weighted Average)}$$

The GIV is the size-weighted outcome minus the equal-weighted outcome:

$$z_t := L_{\Gamma t} = L_{St} - L_{Et} \qquad \text{(GIV)}$$

The GIV isolates purely idiosyncratic firm shocks

Substitute the demand equation into the aggregate definitions:

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$
$$L_{Et} = \phi^d w_t + \eta_t + u_{Et}$$

The common components $(\phi^d w_t + \eta_t)$ difference out:

$$z_t = u_{\Gamma t} = u_{St} - u_{Et} = \underbrace{\sum S_i L_{it}}_{\text{Size-Weighted}} - \underbrace{\frac{1}{N}\sum L_{it}}_{\text{Equal-Weighted}}$$

z_t is made *only* of idiosyncratic shocks.

The GIV yields both supply and demand elasticities

By exogeneity, $\mathbb{E}[u_t \varepsilon_t] = 0 \implies \mathbb{E}[z_t \varepsilon_t] = 0$, where $u_t = (u_{it})_{i=1,\ldots,N}$.

Using the supply equation (38), $\mathbb{E}[(w_t - \psi L_{St}) z_t] = 0$:

$$\psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]} \qquad \text{(Supply Elasticity)}$$

Using the equal-weighted demand equation, $\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0$:

$$\phi^d = \frac{\mathbb{E}[L_{Et} z_t]}{\mathbb{E}[w_t z_t]} \qquad \text{(Demand Elasticity)}$$

The Full Model

Handling heterogeneous factor loadings

Superstar firms respond differently to macro trends

The simple GIV $(L_{St} - L_{Et})$ assumes every firm has the same loading $\lambda_i = 1$. If large firms are more sensitive to the Fed, the simple GIV is **"leaky"**—it correlates with η_t.

The Full Factor Model for granular firms

We now allow for heterogeneous loadings λ_i:

$$L_{it} = \phi w_t + \lambda_i \eta_t + u_{it}$$

The simple GIV becomes:

$$z_t = (\lambda_S - \lambda_E)\eta_t + (u_{St} - u_{Et})$$

If $\lambda_S \neq \lambda_E$, the instrument is **invalid** ($\mathbb{E}[z_t \eta_t] \neq 0$).

The Solution: Orthogonal Optimal Weights Γ

We need weights Γ such that the factor components **cancel exactly**.

$$\sum_{i=1}^{N} \Gamma_i \lambda_i = 0 \quad \text{(Orthogonality Condition)}$$

This ensures $z_t = \sum \Gamma_i u_{it}$ is purely idiosyncratic.

Deriving Γ: Orthogonal Projection

Γ is the "pure granular weight", orthogonal to macro-sensitivities.

We want weights Γ_i that maximize power (by staying close to firm sizes S_i) while ensuring orthogonality to the common factors (λ_i) and a constant (to sum to zero).

Optimization Problem:

$$\min_{\Gamma} \sum_{i=1}^{N} (\Gamma_i - S_i)^2 \quad \text{subject to} \quad \sum \Gamma_i = 0, \quad \sum \Gamma_i \lambda_i = 0$$

Solution (Linear Projection):

$$\Gamma = S - \lambda(\lambda'\lambda)^{-1}\lambda' S$$

Γ is simply the residual of regressing firm sizes (S) onto the factor loadings (λ). It removes the macro-contaminated portion of size.

Estimating Latent Factors with PCA

What if the true factor loadings (λ_i) are unobserved?

In practice, we often do not know exactly how each firm responds to every macro trend. We must estimate the common factors (η_t) and their loadings (λ_i) from the data itself.

- **Principal Component Analysis (PCA):** We apply PCA to the panel data of firm outcomes (y_{it}) to extract the largest sources of common variance.
- **The First Principal Components:** The first K principal components represent the dominant macroeconomic factors (η_t).
- **Factor Loadings:** The firm-specific sensitivities to these components become our estimated $\hat{\lambda}_i$.
- **Calculating Γ:** We then project firm size S orthogonal to $\hat{\lambda}$ to find the optimal, bias-free GIV weights (Γ).

Implementation and Assumptions

Identifying wage spillovers from superstar firms

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How do we apply GIVs to labor markets (e.g., tech sector)?

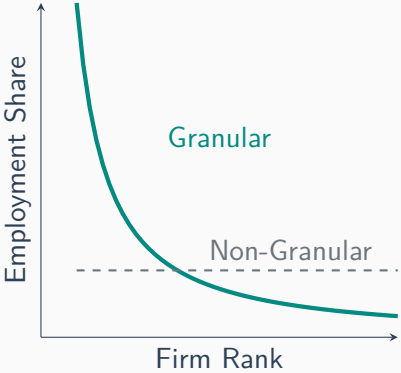
- **The granular shock:** Use idiosyncratic productivity or demand shocks of the largest employers.
- **The instrument:** The GIV (z_t) aggregates these large-firm idiosyncratic shocks.
- **The estimation:** Instrument local or aggregate wage changes (w_t) using the GIV to estimate the causal impact of labor demand on wages (ψ).

GIV precision relies on concentrated labor markets and volatile micro-shocks

For the instrument to be strong, we need:

1. **A large excess Herfindahl index (h):** A few superstar firms must account for a substantial fraction of total employment.

2. **Volatile idiosyncratic shocks (σ_u):** Firm-specific shocks must be large relative to aggregate macroeconomic volatility (σ_ε).



Appendix

Aggregate volatility decays at $1/\sqrt{N}$ under equal weights

Suppose there are N firms, each receiving an independent idiosyncratic shock u_{it}. Assume these shocks have mean 0 and a common variance σ^2:

$$\mathbb{E}[u_{it}] = 0, \quad \text{Var}(u_{it}) = \sigma^2, \quad \text{Cov}(u_{it}, u_{jt}) = 0 \text{ for } i \neq j$$

In a non-granular economy, the aggregate shock is an **equal-weighted** average:

$$u_{Et} = \frac{1}{N} \sum_{i=1}^{N} u_{it}$$

The variance of the aggregate shock is:

$$\text{Var}(u_{Et}) = \text{Var}\left(\frac{1}{N} \sum u_{it}\right) = \frac{1}{N^2} \sum \text{Var}(u_{it}) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}$$

The aggregate volatility is therefore: $\sigma_{agg} = \frac{\sigma}{\sqrt{N}}$